1.13 Algebra: Algebraic Fractions

Worked Solutions

1 a)

$\frac{(x+4)(x+3)}{x+3} = x + 4$

1 b)

$\frac{(x+5)(x+6)}{x+5} = x + 6$

1 c)

$\frac{x(x+5)}{(x+2)(x+5)} = \frac{x}{x+2}$

1 d)

$\frac{(x+7)(x-5)}{(x+7)(x-3)} = \frac{x-5}{x-3}$

1 e)

$\frac{(x+1)(2x-1)}{(2x-1)(x+8)} = \frac{x+1}{x+8}$

1 f)

$\frac{(3x+4)(x+1)}{(3x-4)(3x+4)} = \frac{x+1}{3x-4}$

1 g)

$\frac{(2x+1)(x+4)}{x(2x+1)(2x-5)} = \frac{x+4}{x(2x-5)}$

1 h)

$\frac{(2x+5)(x+6)}{x(2x^2+11x+15)} = \frac{(2x+5)(x+6)}{x(2x+5)(x+3)} = \frac{x+6}{x(x+3)}$

2 a)

$\frac{1}{x} + \frac{3}{2x} = \frac{2}{2x} + \frac{3}{2x} = \frac{5}{2x} = \frac{5}{2x}$

2 b)

$\frac{1}{x^2} + \frac{1}{x} + \frac{3}{5} = \frac{5}{5x^2} + \frac{5x}{5x^2} + \frac{3x^2}{5x^2} = \frac{3x^2+5x+5}{5x^2}$

2 c)

$\frac{1}{x} + \frac{1}{x+1} = \frac{x+1}{x(x+1)} + \frac{x}{x(x+1)} = \frac{2x+1}{x(x+1)}$

2 d)

$\frac{1}{x+2} + \frac{2}{x+4} = \frac{x+4}{(x+2)(x+4)} + \frac{2(x+2)}{(x+2)(x+4)} = \frac{x+4+2x+4}{(x+2)(x+4)}$

$= \frac{3x+8}{(x+2)(x+4)}$

2 e)

$\frac{3}{x+1} - \frac{1}{x-3} = \frac{3(x-3)}{(x+1)(x-3)} - \frac{x+1}{(x+1)(x-3)} = \frac{3x-9-(x+1)}{(x+1)(x-3)}$

$= \frac{2x-10}{(x+1)(x-3)} = \frac{2(x-5)}{(x+1)(x-3)}$

2 f)

$\frac{5}{x+2} + \frac{7}{2x-1} = \frac{5(2x-1)}{(2x-1)(x+2)} + \frac{7(x+2)}{(2x-1)(x+2)}$

$= \frac{5(2x-1)+7(x+2)}{(2x-1)(x+2)} = \frac{10x-5+7x+14}{(2x-1)(x+2)} = \frac{17x+9}{(2x-1)(x+2)}$

2 g)

$\frac{3}{2(x+5)} + \frac{1}{4(x-3)} = \frac{6(x-3)}{4(x+5)(x-3)} + \frac{x+5}{4(x+5)(x-3)}$

$\frac{6x-18+x+5}{4(x+5)(x-3)} = \frac{7x-13}{4(x+5)(x-3)}$

2 h)

$$\frac{x}{x^2-4} + \frac{5}{x+2} = \frac{x}{(x+2)(x-2)} + \frac{5(x-2)}{(x+2)(x-2)}$$
$$= \frac{x+5x-10}{(x+2)(x-2)} = \frac{6x-10}{(x+2)(x-2)} = \frac{2(3x-5)}{(x+2)(x-2)}$$

3 a)

$\frac{(x-1)(x-4)}{x(x-1)} \times \frac{2x+3}{5(x-4)} = \frac{2x+3}{5x}$

3 b)

$\frac{x-3}{3x(x-2)} \times \frac{3(x^2-4)}{(x-3)(x+3)} = \frac{1}{x(x-2)} \times \frac{(x-2)(x+2)}{x+3} = \frac{x+2}{x(x+3)}$

3 c)

$\frac{4(x^2-1)}{3x-1} \times \frac{(3x-1)(x+5)}{2(x+1)} = \frac{4(x+1)(x-1)}{1} \times \frac{x+5}{2(x+1)}$

$= \frac{2(x-1)}{1} \times \frac{x+5}{1} = 2(x-1)(x+5)$

3 d)

$$\frac{x+2}{x+3} \div \frac{(x+2)(x+1)}{2(x^2-x-6)} = \frac{x+2}{x+3} \div \frac{(x+2)(x+1)}{2(x-3)(x+2)}$$
$$= \frac{x+2}{x+3} \times \frac{2(x-3)(x+2)}{(x+2)(x+1)}$$
$$= \frac{x+2}{x+3} \times \frac{2(x-3)}{x+1} = \frac{2(x-3)(x+2)}{(x+3)(x+1)}$$

4 a)

$\frac{1}{x} \times \frac{x+1}{1} = \frac{x+1}{x}$

4 b)

$\frac{1}{x-2} \times \frac{2x}{1} = \frac{2x}{x-2}$

4 c)

$\frac{1+\frac{x+3}{x-5}}{x-5} = \frac{x-5+x+3}{(x-5)^2} = \frac{2x-2}{(x-5)^2} = \frac{2(x-1)}{(x-5)^2}$

4 d)

$\frac{\frac{x-1}{x+4}+1}{x+2} = \frac{x-1+(x+4)}{(x+2)(x+4)} = \frac{2x+3}{(x+2)(x+4)}$

4 e)

$\frac{(3x+1)^2(5x-1)+4(5x-1)(3x+1)}{5x-1} = \frac{(3x+1)^2+4(3x+1)}{1}$

$= 9x^2 + 6x + 1 + 12x + 4 = 9x^2 + 18x + 5$

$= (3x + 5)(3x + 1)$

5 a)

$\frac{3}{x+2} + \frac{x-2}{(x+2)(x+3)} = \frac{3(x+3)}{(x+2)(x+3)} + \frac{x-2}{(x+2)(x+3)}$

$= \frac{3(x+3)+x-2}{(x+2)(x+3)} = \frac{3x+9+x-2}{(x+2)(x+3)} = \frac{4x+7}{(x+2)(x+3)}$ as req.

5 b)

$\frac{(2x+1)(x-1)}{(x+1)(x-1)} - \frac{1}{x(x+1)} = \frac{x(2x+1)}{x(x+1)} - \frac{1}{x(x+1)} = \frac{2x^2+x-1}{x(x+1)}$

$= \frac{(2x-1)(x+1)}{x(x+1)} = \frac{2x-1}{x}$ as req.

5 c)

$\frac{(x-3)(x+3)}{(x-3)(x+3)} - \frac{x+3}{(x-3)(x+3)} + \frac{2}{(x-3)(x+3)}$

$= \frac{(x-3)(x+3)-(x+3)+2}{(x-3)(x+3)}$

$= \frac{x^2-9-x-3+2}{(x-3)(x+3)}$

$= \frac{x^2-x-10}{2(x-1)(x+3)}$ as req.

5 d)

$\frac{(9-4x)(x+3)}{2(x-1)(x+3)} + \frac{2(x-1)}{2(x-1)(x+3)} + \frac{4(x-1)(x+3)}{2(x-1)(x+3)}$

$= \frac{(9-4x)(x+3)+2(x-1)+4(x-1)(x+3)}{2(x-1)(x+3)}$

$= \frac{-4x^2-3x+27+2x-2+4x^2+8x-12}{2(x-1)(x+3)} = \frac{7x+13}{2(x-1)(x+3)}$ as req.